

PHY226 Orbital Motion

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1 Checkpoint1

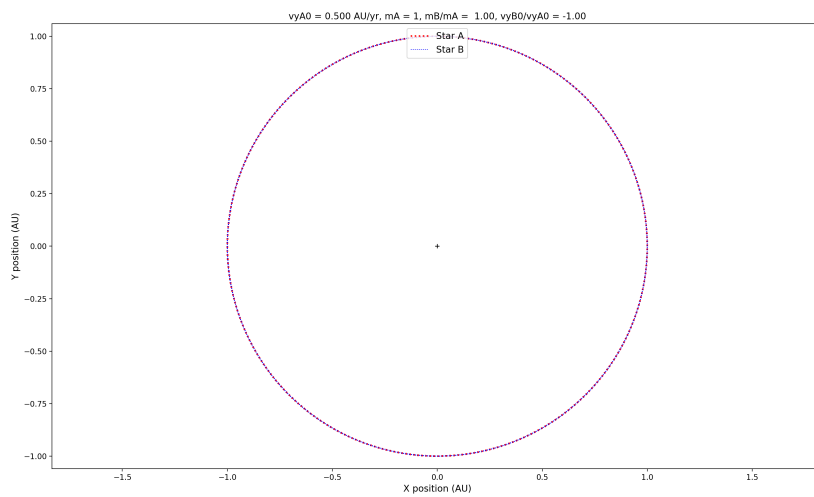


Figure 1: Default

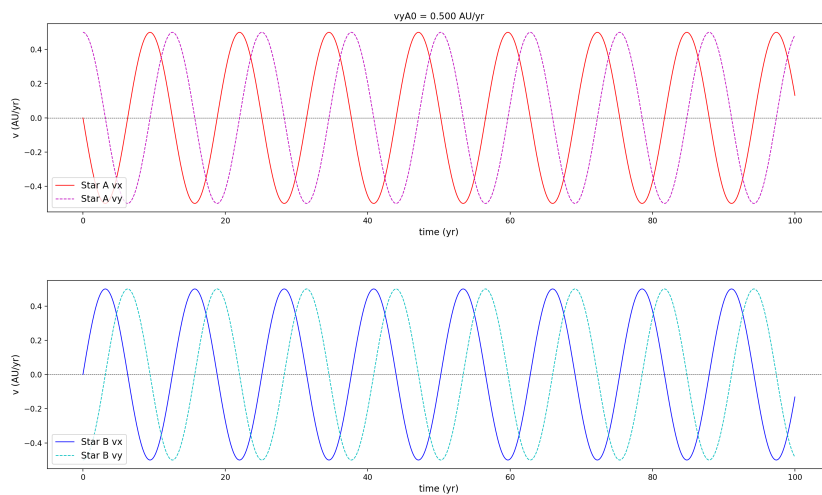


Figure 2: Default

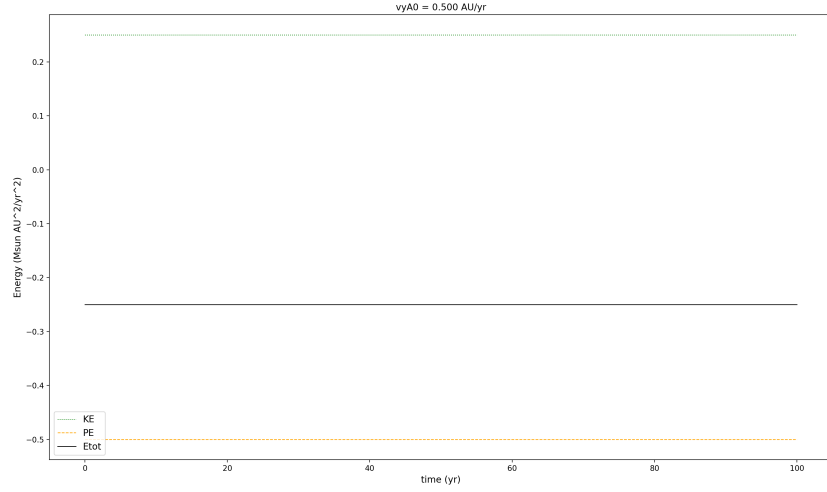


Figure 3: Default

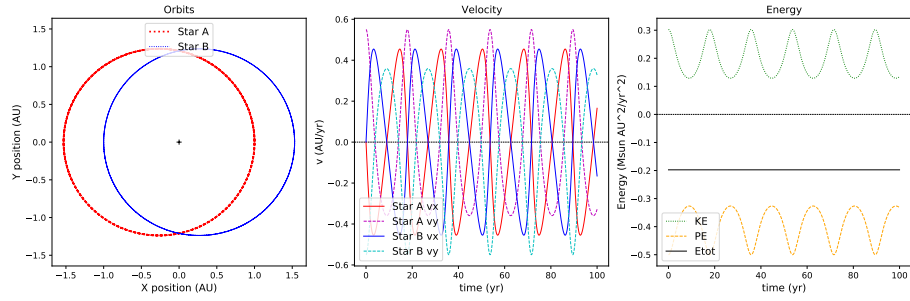


Figure 4: $vyA0=0.55$

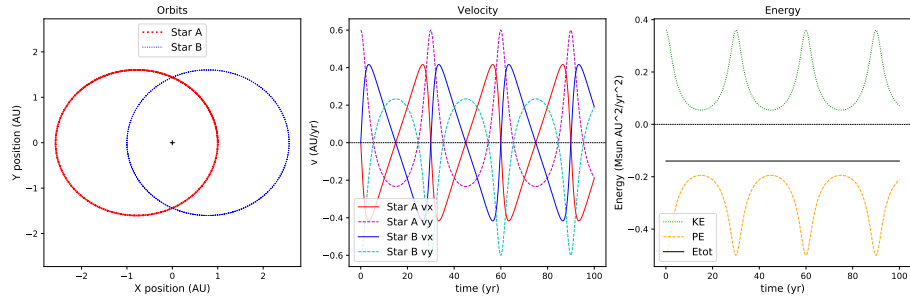


Figure 5: $vyA0=0.6$

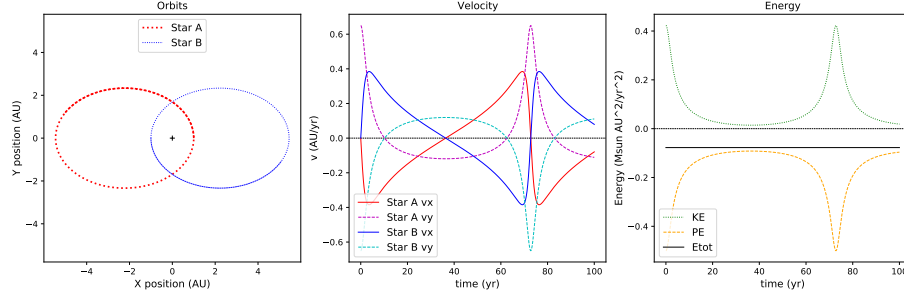


Figure 6: $v_{yA0}=0.65$

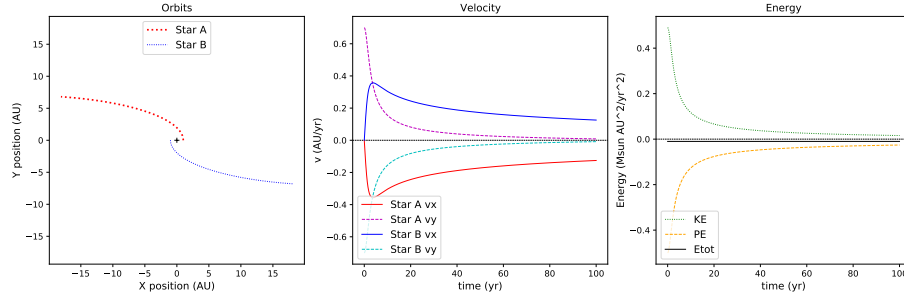


Figure 7: $v_{yA0}=0.7$

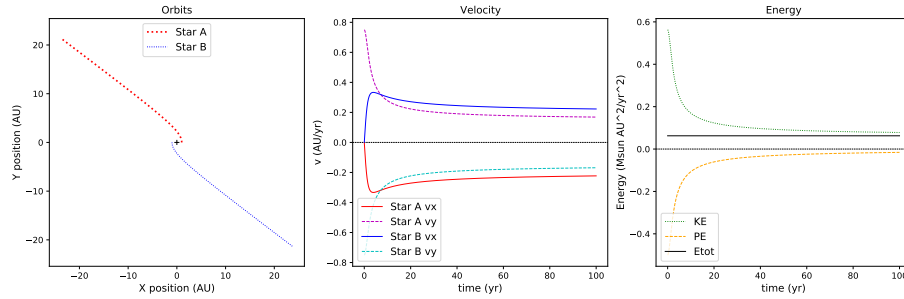


Figure 8: $v_{yA0}=0.75$

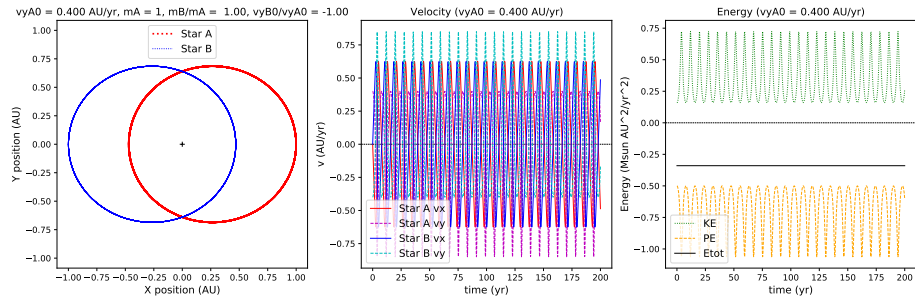


Figure 9: $v_{yA0}=0.4$. The system is bounded since the total energy is negative.

2 Checkpoint2

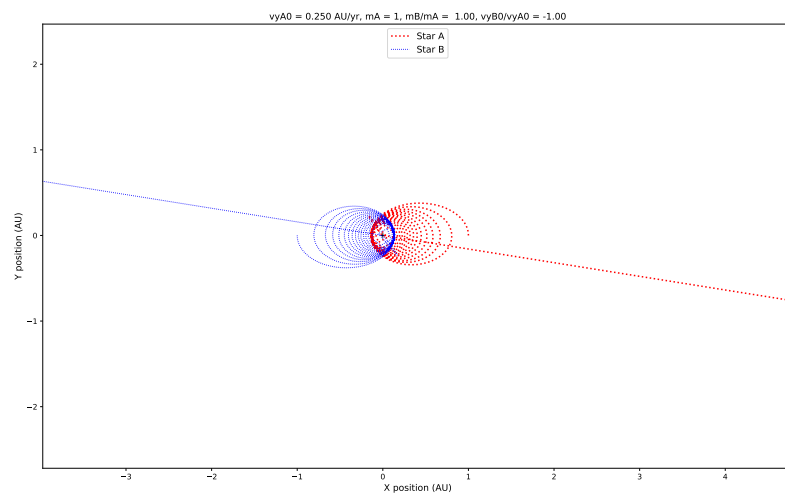


Figure 10: Unbounded $v_{yA0}=0.25$

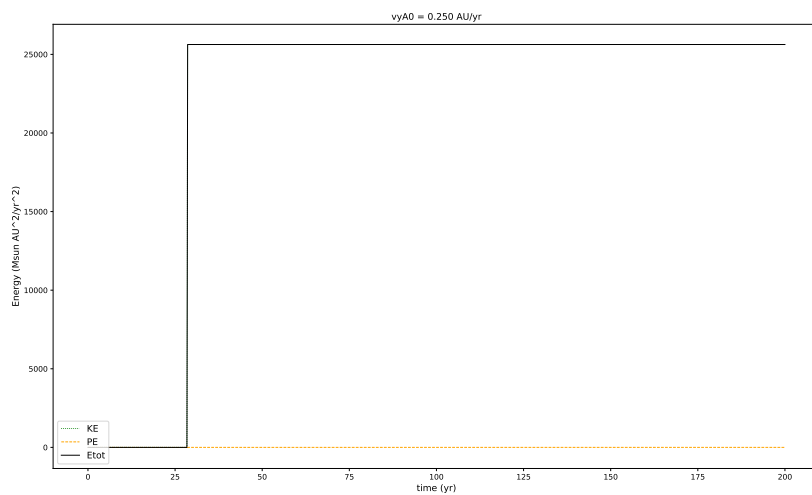


Figure 11: Unbounded, positive Energy

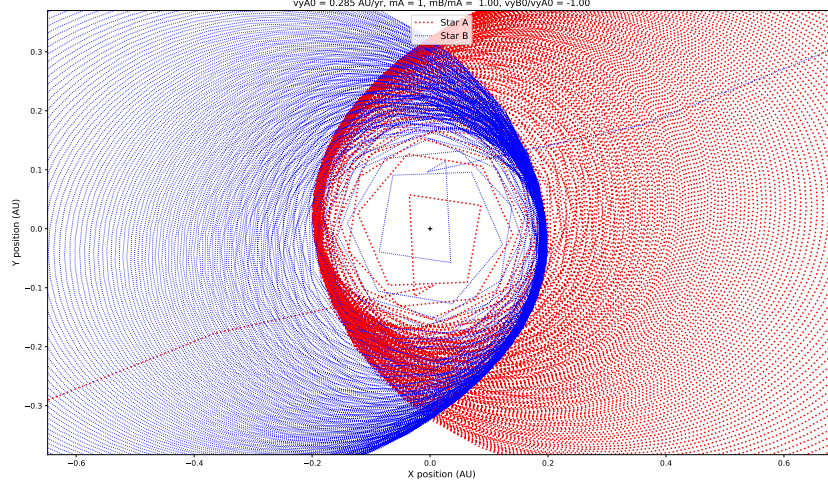


Figure 12: $a=0.324$

3 Checkpoint3

$$\tau_{cross} = \frac{P}{2\pi} = \frac{0.75}{2\pi} = 0.12$$

$$P^2 = \frac{4\pi^2}{GM_{tot}} a^3 = \frac{4\pi^2}{2} (0.324)^3 = 0.6713 \rightarrow P = 0.81$$

Equation 1 gives $P = 0.81$. and the P measured is the **0.75** year. They are similar as expected. And the

$$\tau_{cross} = \frac{P}{2\pi} = \frac{0.75}{2\pi} = 0.12 \approx dt = 0.1$$

4 Exploration

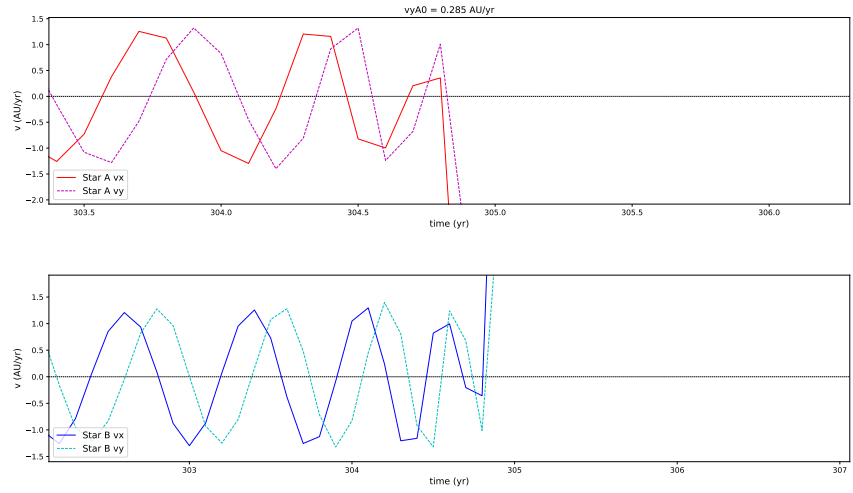


Figure 13: Period $P=0.75$

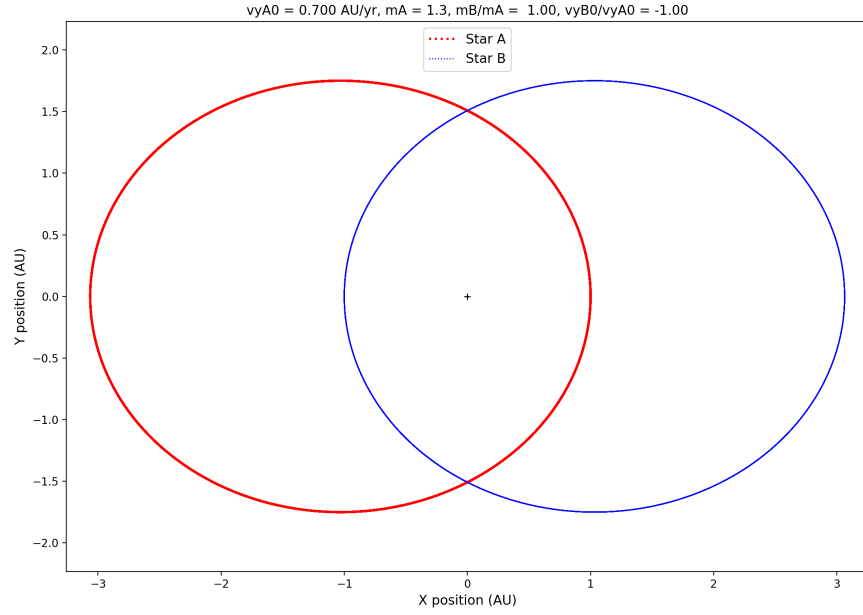


Figure 14: Change mA to $1.3mA$

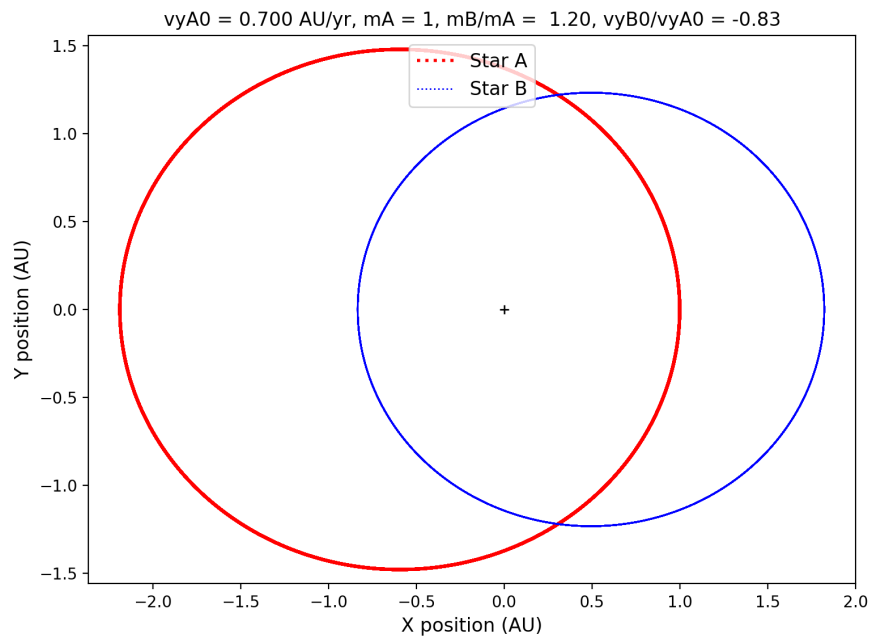


Figure 15: Change $m_B:m_A$ ratio

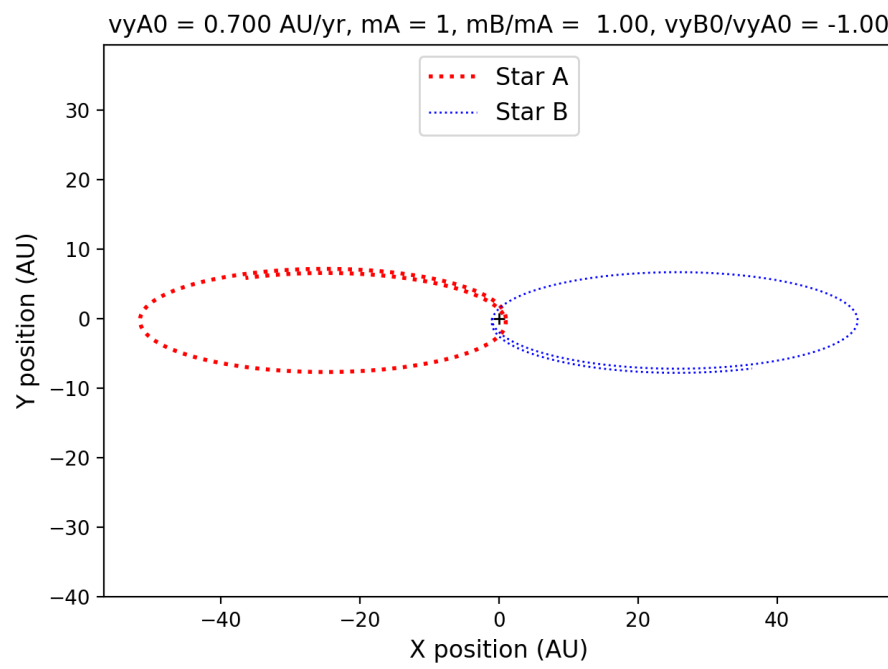


Figure 16: Change v_{yB0}/v_{yA0} , -1.001

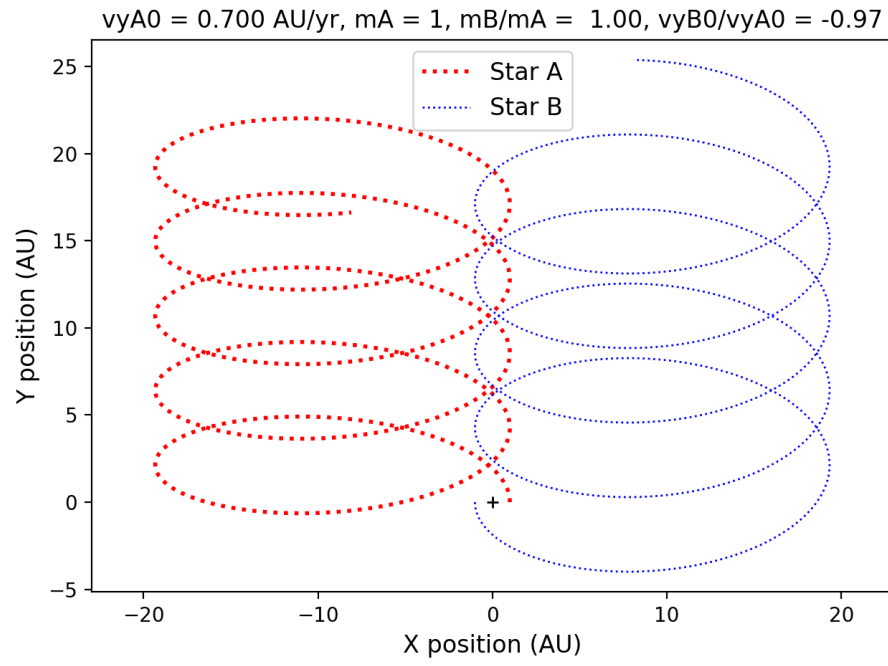


Figure 17: Change v_{yB0}/v_{yA0} , -0.97